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High hypoxia status in pancreatic cancer is associated with multiple hallmarks of an immunosuppressive tumor microenvironment

Hassan Sadozai, Animesh Acharjee, [Hateem Z. Kayani](#), Thomas Gruber, Reginald M. Gorczynski, [Bernard Burke](#)

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Abstract

Introduction: Pancreatic ductal adenocarcinoma (PDAC), the most common form of pancreatic cancer, is a particularly lethal disease that is often diagnosed late and is refractory to most forms of treatment. Tumour hypoxia

is a key hallmark of PDAC and is purported to contribute to multiple facets of disease progression such as treatment resistance, increased invasiveness, metabolic reprogramming, and immunosuppression.

Methods: We used the Buffa gene signature as a hypoxia score to profile transcriptomics datasets from PDAC cases. We performed cell-type deconvolution and gene expression profiling approaches to compare the immunological phenotypes of cases with low and high hypoxia scores. We further supported our findings by qPCR analyses in PDAC cell lines cultured in hypoxic conditions.

Results: First, we demonstrated that this hypoxia score is associated with increased tumour grade and reduced survival suggesting that this score is correlated to disease progression. Subsequently, we compared the immune phenotypes of cases with high versus low hypoxia score expression (HypoxiaHI vs. HypoxiaLOW) to show that high hypoxia is associated with reduced levels of T cells, NK cells and dendritic cells (DC), including the crucial cDC1 subset. Concomitantly, immune-related gene expression profiling revealed that compared to HypoxiaLOW tumours, mRNA levels for multiple immunosuppressive molecules were notably elevated in HypoxiaHI cases. Using a Random Forest machine learning approach for variable selection, we identified LGALS3 (Galectin-3) as the top gene associated with high hypoxia status and confirmed its expression in hypoxic PDAC cell lines.

Discussion: In summary, we demonstrated novel associations between hypoxia and multiple immunosuppressive mediators in PDAC, highlighting avenues for improving PDAC immunotherapy by targeting these immune molecules in combination with hypoxia-targeted drugs.

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Keywords

- hypoxia
- tumor microenvironment (TME)
- pancreatic ductal adenocarcinoma (PDAC)
- immune checkpoint
- galectins

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KW - tumor microenvironment (TME)

KW - pancreatic ductal adenocarcinoma (PDAC)

KW - immune checkpoint

KW - galectins

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